Q We are considering IVF. How successful is it and what does it involve?

The live birth rate from IVF is increasing gradually. In 2012, US success rates were 32.8 percent for women under 35; 27.3 percent for 35–37 years; 20.7 percent for 38–39 years; 13.1 percent for 40–42 years; and 4.4 percent for 43 years plus.

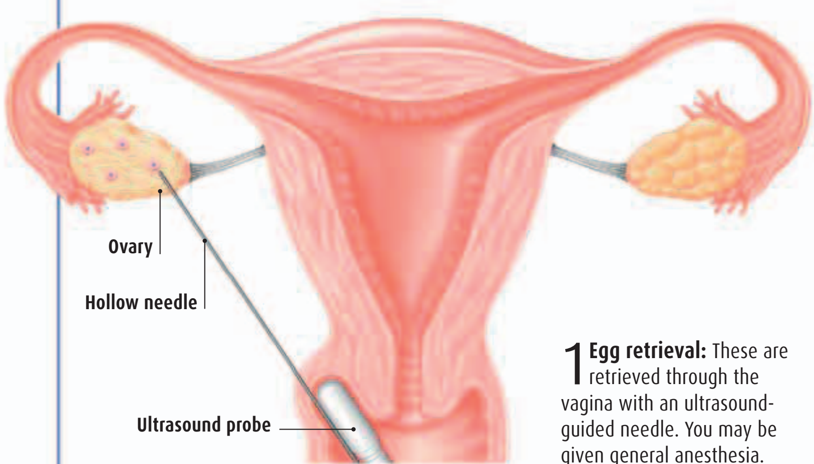
1 in 50
babies **in the US** are **born as a result of IVF treatment** (2014).

IVF is the process by which an egg and sperm are mixed together in a petri dish outside the body. Following fertilization, an embryo (possibly more than one) is transferred to the uterus. The process is extremely time-consuming and involves many appointments. You are also required to take medication by injection and undergo a minor surgical procedure. Before beginning IVF, it is crucial to prepare yourself for the fact that it can be an extremely fraught and draining experience. It can put an enormous amount of pressure on a couple and it is vital, therefore, that both partners are in complete agreement about the course of action.

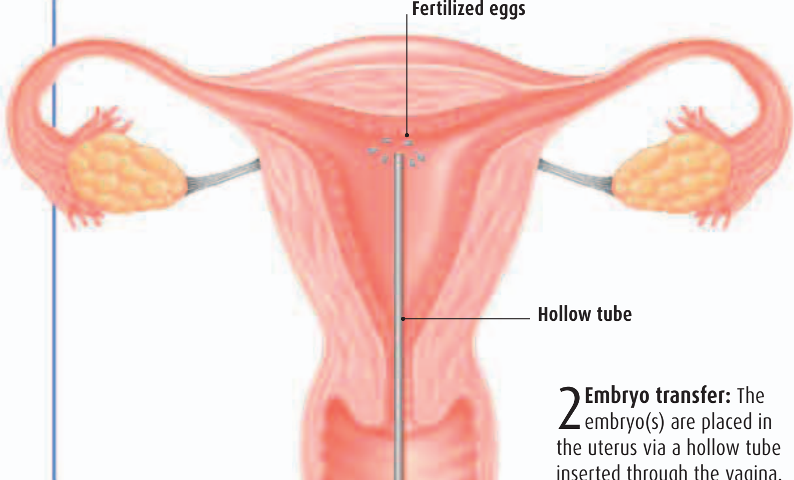
Before embarking on the treatment, speak to your doctor and consider doing your own research into hospitals, clinics, and the procedure. IVF is appropriate in some medical situations, but not in others. There are also some couples for whom IVF is likely to be of little help, so a full understanding of what it involves is essential.

WHAT HAPPENS WHEN?

A number of eggs are usually required to increase the likelihood of conceiving a viable embryo. IVF treatment involves injections to encourage the ovaries to ripen more than one egg. Your progress is monitored with scans and a specialist will choose an optimum point to collect the eggs for fertilization.



1 **Egg retrieval:** These are retrieved through the vagina with an ultrasound-guided needle. You may be given general anesthesia.



2 **Embryo transfer:** The embryo(s) are placed in the uterus via a hollow tube inserted through the vagina.

THE STAGES OF IVF

Stage 1–Suppression
In some treatments, the ovaries are suppressed before stimulating them to produce multiple eggs. This stage can begin one week before your period is due or the day after it begins. You take a nasal spray or inject a drug for one to two weeks to suppress your natural cycle. This prevents the pituitary gland from producing follicle-stimulating hormones (FSH). Other treatments go straight to stimulation (see below) and suppress the release of mature eggs afterward.

Stage 2–Stimulation
An ultrasound scan is done to confirm that the ovaries are not active. Following this, you take FSH as a daily injection for 10 to 12 days. This increases the number of eggs your ovaries produce. During this time regular hormone tests are done to indicate how well the follicles are responding to the FSH. A last injection given approximately 36 hours before egg collection helps the eggs to mature.

Stage 3–Egg retrieval
This is usually done using a vaginal ultrasound that shows an image of each ovary. The eggs are sucked into a test tube and given to an embryologist who places them in special fluid to be examined.

Stage 4–Embryo selection
The male partner produces semen and his sperm are mixed with the eggs and placed in an incubator. The embryos are assessed 48 hours later and checked to ensure they look normal. The ideal time to transfer the embryo(s) is after about five days when it has developed into a blastocyst. This gives a higher chance of pregnancy.

Stage 5–Embryo transfer
An embryo (or embryos) is put into a fine plastic tube and this is inserted through the cervix into the uterus.